

Multiple Choice (20 marks)

1. Which of the following memories uses one transistor and one capacitor as basic memory unit
 - (a) SRAM
 - (b) DRAM
 - (c) Both SRAM and DRAM
 - (d) None
2. In a lap winding dc machine number of conductors are 100 and number of parallel paths are 10. Find the average pitch
 - (a) 10
 - (b) 100
 - (c) 50
 - (d) 1
3. While operating on variable frequency supplies, the AC motor requires variable voltage in order to
 - (a) protect the insulation.
 - (b) avoid effect of saturation.
 - (c) improve the capabilities of the inverter.
 - (d) protect thyristor from dV / dt .
4. One of the following is the primary function of an oscillator
 - (a) produces sinusoidal oscillations
 - (b) generates non sinusoidal waveforms
 - (c) generates sustained oscillations at a constant amplitude and specific frequency
 - (d) none of the above
5. _____ is the most detrimental impurity in the magnetic materials
 - (a) Carbon.
 - (b) Sulphur.
 - (c) Oxygen.
 - (d) Nitrogen.
6. The resistivity of the conductor depends on

- (a) area of the conductor.
 - (b) length of the conductor.
 - (c) type of material.
 - (d) none of these.
7. If all the elements in a particular network are linear, then the superposition theorem would hold, when the excitation is
- (a) DC only
 - (b) AC only
 - (c) Either AC or DC
 - (d) An Impulse
8. According to the Bohr model, an electron gains or losses energy only by
- (a) moving faster or slower in an allowed orbit.
 - (b) jumping from one allowed orbit to another.
 - (c) being completely removed from an atom.
 - (d) jumping from one atom to another atom.
9. Electrons in p-type material of a semi-conductor are called as
- (a) either minority carriers or majority carriers
 - (b) minority carriers
 - (c) majority carriers
 - (d) valance carriers
10. Which oscillators are easy to fabricate in a monolithic IC?
- (a) Crystal oscillator.
 - (b) Hartley oscillator.
 - (c) Wien bridge oscillator.
 - (d) Relaxation oscillator.

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Lowest critical frequency is due to pole and it may be present at the origin or nearer to the origin, then the type of network is' is 'LC.'.

2. True or False: The correct answer to 'Purely mechanical instrument cannot be used for dynamic measurements because they have' is 'all of the above.'
3. True or False: The correct answer to 'An SCR has half cycle surge current rating of 3000 A for 50 Hz supply. One cycle surge current will be' is '6000 A.'
4. True or False: The correct answer to 'For a dc machine shunt resistance and armature resistance values are' is 'high and low'.
5. True or False: The correct answer to 'For audio frequency applications, the popular oscillator used is' is 'Wien bridge oscillator'.
6. True or False: The correct answer to 'The normal voltage used in directional heating is' is '66 KV.'
7. True or False: The correct answer to 'In AC circuits, the connection of measuring instruments cause loading effect errors which may effect' is 'only phase of the quantity being measured.'
8. True or False: The correct answer to 'Which of the following is a digital device' is 'Microphone'.
9. True or False: The correct answer to 'Stability of a transmission line can be increased by' is 'both A and B'.
10. True or False: The correct answer to 'A Binary number system has how many digits.' is '1'.

End of Paper